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Project Title:	Systems Biology Model of Interactions between Tissue Growth Factors and DNA Damage Pathways: Low Dose Response and Cross-Talk in TGFBeta and ATM Signaling		
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Task Description:

The etiology of radiation carcinogenesis has been described in terms of aberrant changes that span several levels of biological organization. Growth factors regulate many important cellular and tissue functions including apoptosis, differentiation, and proliferation. A variety of genetic and epigenetic changes of growth factors have been shown to contribute to cancer initiation and progression. It is known that cellular and tissue damage to ionizing radiation is in part initiated by the production of reactive oxygen species, which can activate cytokine signaling, and the DNA damage response pathways, most notably the ATM signaling pathway. Recently the transforming growth factor beta pathway has been shown to regulate or directly interact with the ATM pathway in the response to radiation. The relevance of this interaction with the ATM pathway is not known although p53 becomes phosphorylated and DNA damage responses are involved. However, growth factor interactions with DNA damage responses have not been elucidated particularly at low doses and further characterization of their relationship to cancer processes is warranted.
Our goal will be to use a systems biology approach to mathematically and experimentally describe the low dose responses and cross-talk between the ATM and TGF beta pathways initiated by low and high LET radiation. We will characterize ATM and TGF beta signaling in epithelial and fibroblast cells using 2D models and compare these data to signaling patterns in physiologically relevant 3D organotypic cell culture models to elucidate possible cell type and context specific differences in inter-cellular communication. We will investigate the roles of the Smad and Activating transcription factor 2 (AFT2) proteins as the potential major contributors to crosstalk between the TGF beta and ATM pathways, and links to cell cycle control and/or the DNA damage response, and potential differences in their responses at low and high doses. We have developed various experimental approaches to apply to these problems, including confocal microscopy and flow cytometry, to detail changes at low dose/dose-rate in order to understand individual cell responses, and will establish our mathematical models based on the experimental findings resulting from change in DNA repair, apoptosis, and proliferation.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

Changes in growth factor signaling due to either genetic and/or epigenetic alterations have been shown to play a role in cancer initiation and progression. Transforming growth factor beta (TGFß) is rapidly and persistently activated in response to DNA damage. It has an important role in controlling cellular proliferation, and differentiation among other cellular functions. Cellular and tissue damage following ionizing radiation is in part initiated by the production of reactive oxygen species (ROS) that activate cytokine signaling, and the DNA damage response pathways, including the ATM signaling pathway. It has been proposed that radiation induced stimulation of ATM and its downstream components involves activation of TGFß-1 and may result from TGFß-1-receptor I-Smad signaling. It was suggested that perhaps TGFß signaling primes cells to respond to DNA damage either by helping to recruit ATM to the site of DNA damage, or by activating ATM. Further work confirmed that TGFß-1 had a role in stimulating ATM- and p53 following radiation, but revealed that rapid ATM stimulation was not dependent on TGFß-1-receptor I-Smad signaling. Thus the relationship between TGFß and ATM pathways is complex and the cross-talk between these pathways may differ dependent on the dose or radiation quality of the exposure, which is something we are investigating in our studies. For these studies we have focused primarily on esophageal cells, using both 2D and 3D culture systems, as the human esophagus is an organ sensitive to radiation exposure. Experiments were designed to characterize ATM and TGFß signaling, as well as the interrelationship of each pathway's signaling by various means. Responses were monitored both in epithelial and fibroblast esophageal cells, to better understand cell type differences in response. How differences in growth conditions may uniquely influence response was also studied by comparing signaling patterns observed in 2D monolayers to a more physiologically relevant 3D organotypic cell culture model. Studies also investigated the roles of the Smad and Activating transcription factor 2 (AFT2) proteins as potential major nodes for crosstalk between the TGFß and ATM pathways. In these studies we have investigated how cross-talk may influence cell cycle control and/or the DNA damage response, and investigated potential differences in responses at low versus high doses. Both confocal microscopy and flow cytometry are used to study the responses, to detail changes at low dose/dose-rate to understand individual cell responses. Our goal is to use a systems biology approach to mathematically and experimentally describe the low dose responses and cross-talk between the ATM and TGFß pathways initiated by low versus high LET radiation.

Materials & Methods:

2D Monolayer Cultures: Human hTERT immortalized Esophageal cells (EPC), primary human epithelial cells (HEEC), and esophageal fibroblast cells (FEF) were obtained courtesy of Dr. Rustgi (University of Pennsylvania). Cells were cultured in flasks or chamber slides and irradiated at low (5-10 cGy) and higher doses (50 to 100 cGy) and fixed at multiple time points thereafter. For flow cytometry experiments adherent cells were trypsinized from flasks and fixed as a single cell suspension. Single cell suspensions were then counted and stained with primary and secondary antibodies. Samples are analyzed on a BD FACS calibur acquiring at least 10,000 events for each condition. For immunofluorescence studies, cells are fixed with paraformaldehyde (4%) and stained with appropriate primary and secondary antibodies.

3D Esophageal Organotypic cultures: a collagen gel mixed with human fetal esophageal fibroblasts was grown for 7 days on a specialized culture support and subsequently overlaid with EPC. Two days after plating (day 9), a liquid–air interface was created, by removing the medium on top of the supporting membrane, thus allowing stratification of the squamous epithelium, which resembles the human esophageal squamous epithelium. Samples were irradiated at day 14 with Fe 600 MeV/u, Fe1000 MeV/u or Si173 MeV/u using doses 0, 0.1GY and 2Gy. OTC samples were harvested at 24 hrs or 72 hours post IR. 3D tissues were fixed and processed for paraffin embedding. Paraffin embedded samples were sectioned and stained with various antibodies.

Brief summary of past year's research accomplishments: 2D monolayer cultures were used to investigate if TGFß was expressed at measurable levels in esophageal cells, as has been observed in mammary epithelial cells. No published reports of levels of TGFß in esophageal cells plus or minus radiation exposure were available, so we sought to define this using an established assay and a luciferase reporter cell line. MCF10A mammary epithelial cells were used as a control as levels have been reported for this line. We noted cell type dependent differences in TGFß expression upon X-ray exposure. These results may suggest that different cells regulate TGFß differently in response to radiation. Further studies will be needed to confirm these findings.

Previous studies in mammary epithelial cells have noted that TGFß inhibition results in a failure of pATM or

	downstream phosphorylations to take place following radiation. To define how TGFß addition or inhibition affected pATM pathway signaling in esophageal cells, we exposed cells to TGFß or its inhibitor plus or minus radiation (2Gy X-ray). Differences in the foci characteristics were noted with TGFß inhibitor addition, and we are currently following up on whether these changes are persistent. Others and we have noted that ATF2 is phosphorylated at multiple sites following radiation, and that phosphorylations at the different sites are indicative of different cell fates or cellular responses. For instance, the phospho-site activated in response to DNA damage is Ser 490/498, however the phosphorylation site responding to cellular stress, which activates its transcriptional response is Thr 69/70. Previous work in fibroblasts has indicated that upon damage cells respond with phosphorylation of one or the other phospho-protein sites, and that single cells do not show both phosphorylations. We performed studies to determine if esophageal cells showed this pattern and would like to perform additional studies to discern if there is a radiation quality difference in ATF2 phospho-responses and how TGFß inhibition may uniquely affect this pattern. Flow cytometry studies were performed to quantify cellular ATF2 phosphorylation (Ser 490/98) in various cell types following exposure to Si173 MeV/u radiation. We compared esophageal and skin fibroblasts (82-6) and esophageal epithelial cells. Differences in the kinetics of response were noted. A rather new technology, Luminex by Millipore, will be used to probe the cross-talk between the ATM and TGFß pathways. This technology allows the detection of a large number of phosphorylations or protein expressions in a 96 well format. We have chosen to study a number of phospho-proteins in the MAPkinase pathway under various conditions, designed to understand the cross-talk between the TGFß and ATM pathways. Proteins in this panel include phosphorylated forms of ATF2, c-Jun, JNK, MSK1, p38, STAT1, Hsp27, Erk ½, and MEK1. We plan to have cells unexposed, exposed to TGFß, TGFß inhibitor, two different ATM inhibitor concentrations, or ATM inhibitor plus TGFß, each plus or minus a low (0.1 Gy) or a higher dose (1 Gy) of X-ray. These studies will allow us to better understand potential cross-talk between the pathways and upon optimization with X-ray we will repeat the study with high LET radiation. These studies will also provide potential phospho-protein targets to follow up on to develop flow-based assays, ones in which unique radiation quality differences in pathway cross-talk is evident. We have also spent considerable effort optimizing 3D organotypic growth and characterizing the response to radiation. We have observed some unique changes dependent upon radiation following Fe 1000 MeV/u ion exposure. Future studies will attempt to study radiation quality differences in response and incorporate assays plus and minus TGFß and its inhibitor to compare to studies done in 2D. In summary we have noted some evidence of esophageal specific responses to radiation and a suggestion of cross-talk between the TGFß and ATM signaling pathways in these cells. We plan to further investigate the dose-dependent cross-talk with additional high LET radiation experiments, and define phospho-proteins most informative to study using flow cytometry in a more defined kinetic manner. We have optimized 3D organotypic growth of esophageal cultures. Additional studies are needed to define whether 2D monocultures respond uniquely to radiation as compared to 3D organotypic cultures and are currently in progress.
Bibliography Type:	Description: (Last Updated: 11/01/2023)
Abstracts for Journals and Proceedings	Cucinotta FA, Li Y, Wang M, Pluth JM, O'Neill P. "Models of TGFß-Smad Signaling in the DNA Damage Response following Gamma-Rays and HZE Particle Irradiations." 13th International Workshop on Radiation Damage to DNA, Massachusetts Institute of Technology, Cambridge, MA, 14 – 18 June 2014. 13th International Workshop on Radiation Damage to DNA, Massachusetts Institute of Technology, Cambridge, MA, 14 – 18 June 2014. , Jun-2014